

INDIA.RUNS

Build what next India runs on

Team Name: Solo Participant

Team Leader: D Harini

Problem Statement: AI-Powered Candidate Discovery & Ranking System

Solution Overview

- Built an AI-based candidate ranking system.
- Uses semantic search instead of only keyword matching.
- Combines profile relevance, skills, experience, and behavioral signals.
- Generates ranked candidate recommendations automatically.

Differentiator:

- Understands candidate profile meaning using embeddings.
- Considers recruiter response rate, notice period, GitHub activity, etc.

JD Understanding & Candidate Evaluation

Key Requirements Extracted

- Python
- Embeddings
- Retrieval Systems
- Ranking Systems
- LLMs
- Fine Tuning
- Vector Databases

Candidate Signals

- Semantic relevance
- Skills match
- Years of experience
- Open-to-work status
- Relocation willingness
- Recruiter response rate

Ranking Methodology

Model Used

- SentenceTransformer (all-MiniLM-L6-v2)

Ranking Steps

- Generate JD embedding
- Generate candidate embedding
- Calculate cosine similarity
- Compute skills score
- Compute experience score
- Compute behavior score
- Generate final score

Formula

Final Score =

$0.50 \times \text{Semantic Score} +$

$0.15 \times \text{Experience Score} +$

$0.20 \times \text{Skills Score} +$

$0.15 \times \text{Behavior Score}$

Explainability & Data Validation

- Every candidate receives separate scores.
- Ranking decisions are transparent.
- Missing fields handled safely.
- Invalid profiles receive lower scores.
- No hallucinated candidate information used.

End-to-End Workflow

Job Description



Embedding Generation



Candidate Processing



Semantic Matching



Skills Matching



Behavior Scoring

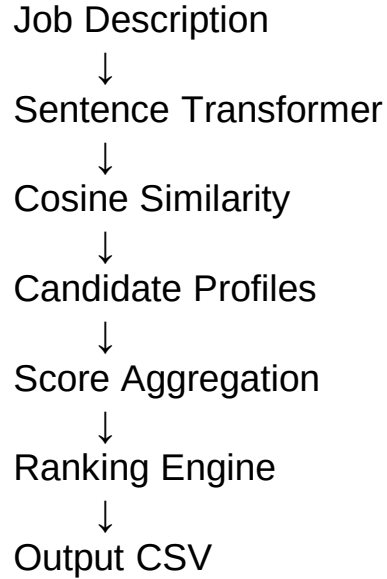


Final Ranking



sample_submission.csv

System Architecture



Results & Performance

- Processed 20,000 candidates.
- Generated ranked candidate list.
- Top candidate score ≈ 0.65 .
- Output generated in required format.

Top Candidate Example

CAND_0018499
Score: 0.6543
Rank: 1

Technologies Used

- Python
- Pandas
- Sentence Transformers
- Scikit-learn
- JSON
- GitHub

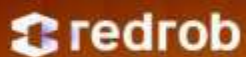
Submission Assets

GitHub Repository

<https://github.com/DasarajuHarini/AI-Candidate-Ranking-System>

Files Submitted

1. advanced_ranking.py
2. rank_candidates.py
3. README.md
4. requirements.txt
5. presentation.pptx
6. AI-Powered Candidate Discovery & Ranking System.pdf
7. sample_submission.csv



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THANK YOU